

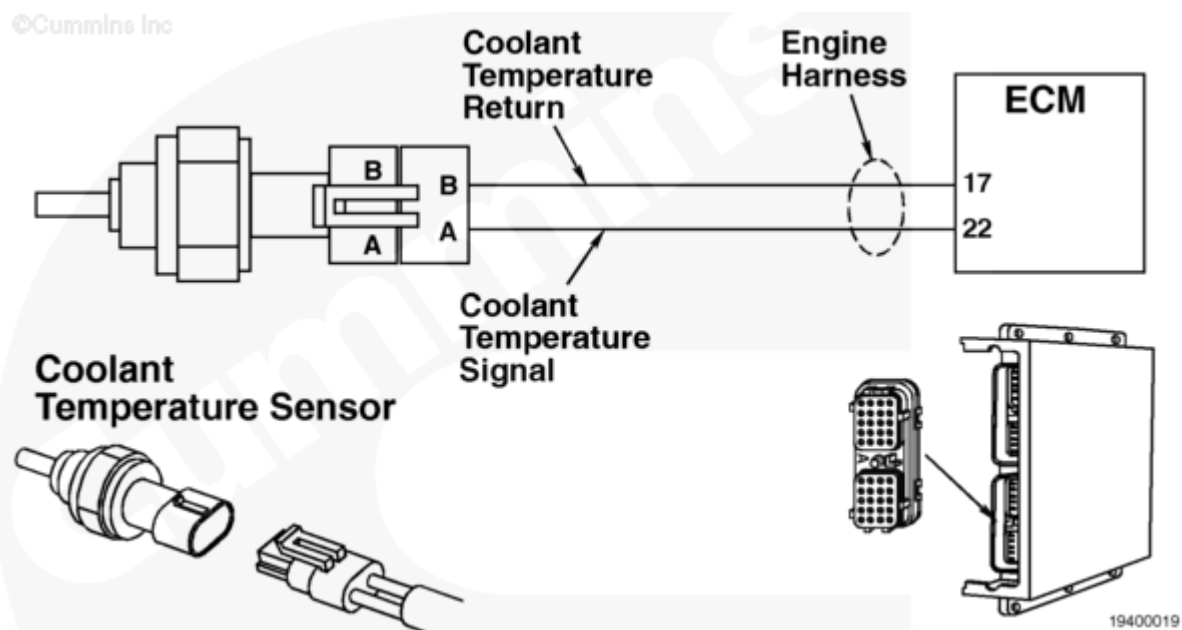
Fault Code: 151

Coolant Temperature - Engine Protection

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Overview

Codes	Reason	Effect
Fault Code: 151 PID(P): P110 SPN: 110 FMI: 0 Lamp: Engine Protection SRT: 00-357	High coolant temperature has been detected. Voltage signal at coolant temperature signal pin 22 indicates the coolant temperature is above 100°C [212°F].	Calibration-dependent progressive power and speed derate and engine shutdown as temperature increases over thresholds. Centinel™ system is disabled.



Coolant Temperature Sensor Circuit

Circuit Description

The coolant temperature sensor is used by the ECM to monitor the temperature of the engine coolant. The coolant temperature is used by the ECM for the engine protection system, timing, and fueling control. The ECM monitors the voltage on pin 22. The ECM expects to see the voltage vary between 0.21 and 4.95 VDC. If the voltage is below 0.21 VDC for more than 2 seconds, then the ECM will log Fault Code 151. Voltage below 0.21 VDC on pin 22 can be caused by a cooling system failure or an in-range sensor failure.

Component Location

QSK19 series engines - The coolant temperature sensor is located on the side of the thermostat housing on most applications.

Shoptalk

- Make sure the airflow through the radiator is **not** obstructed.
- The resistance of all the temperature sensors varies with the temperature. The reading that you observe could possibly compare to the following table if the sensor is functioning properly.

Temperature(°C)	Temperature[°F]	Resistance(ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Refer to Troubleshooting Fault Code t05-151 (</qs3/pubsys2/xml/en/procedures/19/19-t05-151.html>)